
CIS 678 - Machine Learning

– From Data to an Honest Evaluation –

— Instructor: Yong Zhuang —

yong.zhuang@gvsu.edu

Last Class

- Why Learn from Data?
 - Traditional programming and machine learning
- Instance-based and model-based learning
- How Model-Based Learning Works
 - Model selection, training, and prediction

Agenda

- Data challenges,
- Generalization problems,
- Training/validation/test data, and
- Leakage-safe evaluation

Main Challenges of Machine Learning

- Data problems
 - Insufficient Quantity of Training Data
 - Nonrepresentative Training Data
 - Poor-Quality Data
 - Irrelevant Features
- Model problems
 - Overfitting the Training Data
 - Underfitting the Training Data
- Deployment Issues

Insufficient Quantity of Training Data

More representative training data can improve generalization.

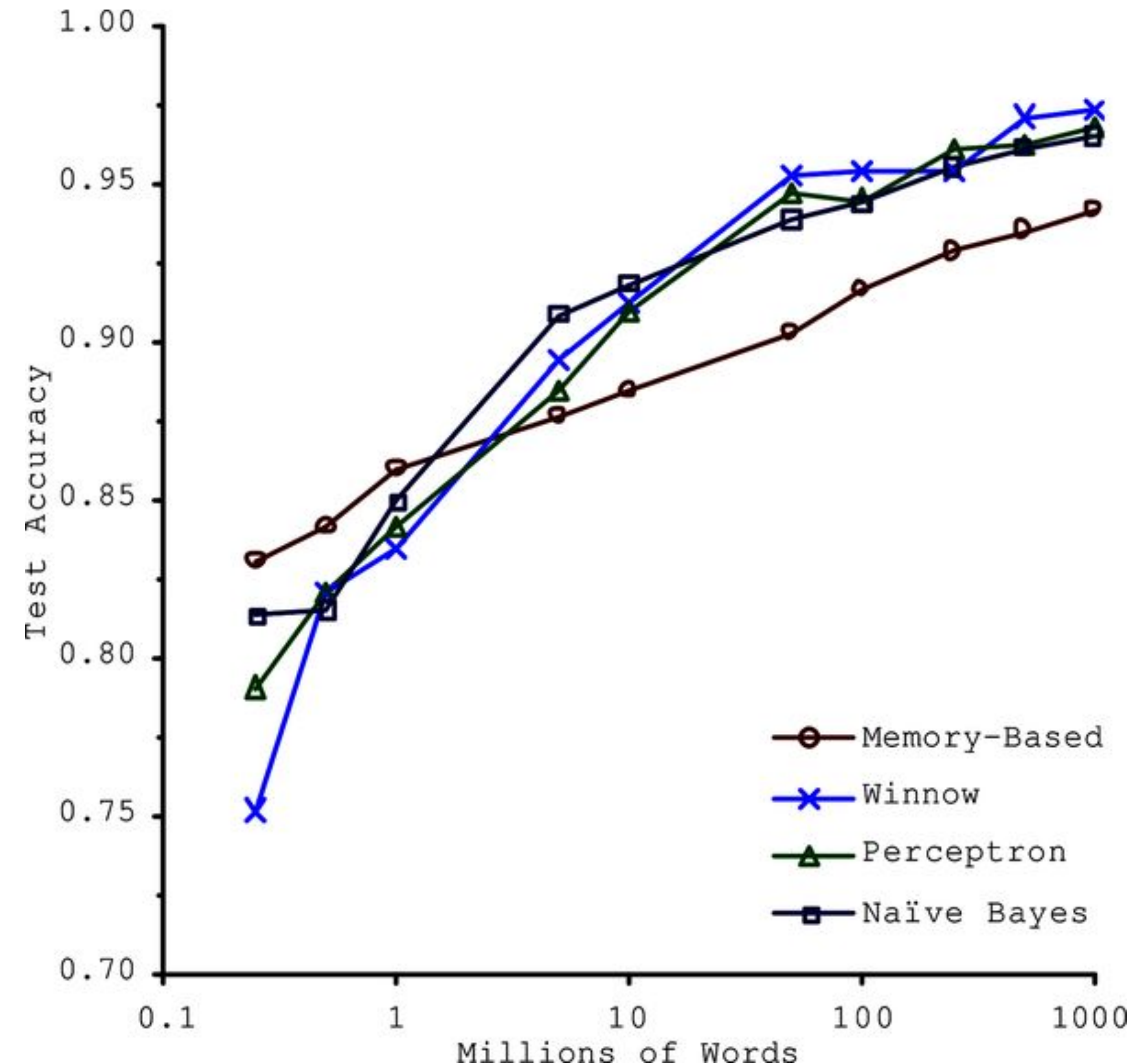
More data often reduces variance.

More data may help a flexible model learn stable patterns.

More data cannot correct biased sampling, incorrect labels, or irrelevant features.



If we increase the training set from 10,000 to 1 million examples, will test performance always improve?



Hands-On Machine Learning with Scikit-Learn and PyTorch - Aurélien Géron

Representative Training Data

Target population

People or cases where the model will make predictions



Data collection process



Training sample

Examples available for model development

Training data should represent the cases the model will encounter after deployment.

- *Are important groups included?*
- *Does the training distribution match the production distribution?*

Sampling Bias

The 1936 Literary Digest Poll

Poll result	Election result
About 10 million questionnaires sent	Franklin Roosevelt won
About 2.4 million responses	Roosevelt received about 62%
Predicted If Landon would win	The prediction was wrong

A large sample can still be biased.

Sampling frame bias: the contact lists overrepresented wealthier voters.

Nonresponse bias: people who responded differed from people who did not.

Poor-Quality Data

Problem	Example
Missing values	A patient has no recorded blood pressure
Incorrect measurements	A faulty sensor reports 500°C
Noise	An image contains random pixel variation
Outliers	One house has an unusually high sale price



Should every outlier be removed?

An unusual value may be an error or a valid rare case.

Feature Selection and Feature Engineering

- Feature selection: retain useful existing features.
- Feature transformation: convert features into a more useful representation.
- Feature construction: create new features from existing data.
- Additional data: collect information that better represents the problem.

House age = Current year - Year built

Generalization

Generalization is performance on data that the model did not use during training.

$$R_{\text{train}}(f) = \frac{1}{n_{\text{train}}} \sum_{i=1}^{n_{\text{train}}} L(f(x_i), y_i)$$

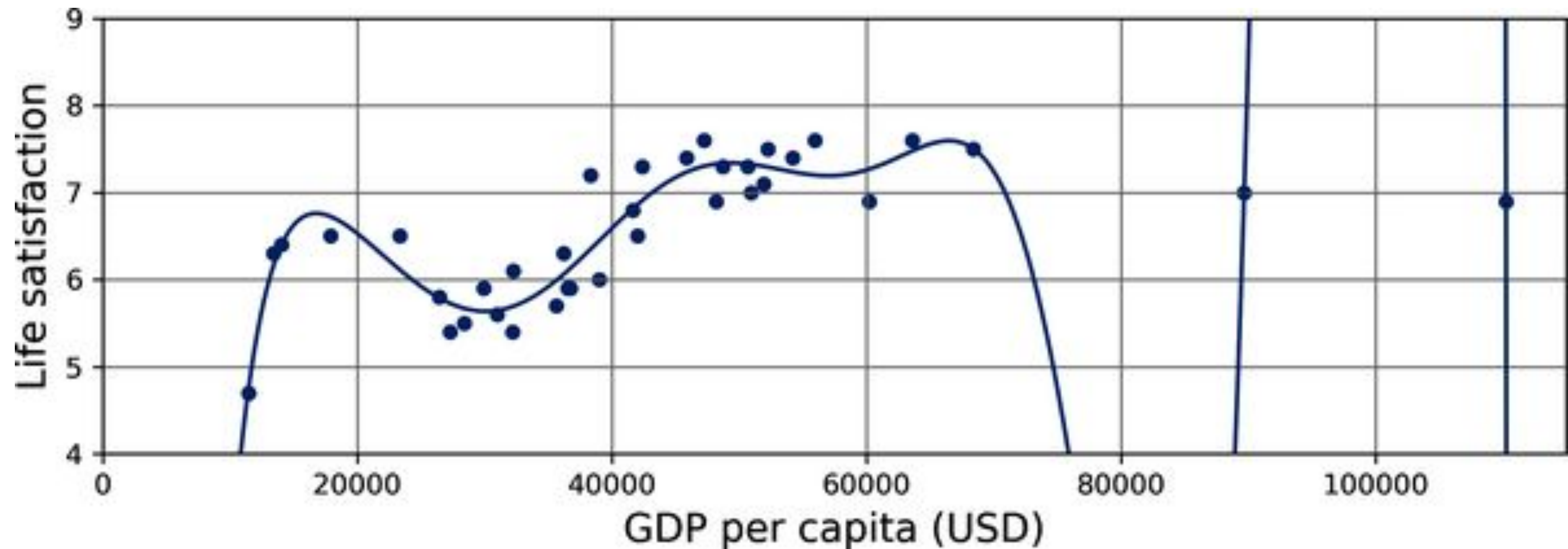
$$R_{\text{test}}(f) = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} L(f(x_i), y_i)$$

$$\text{Generalization gap} = R_{\text{test}}(f) - R_{\text{train}}(f)$$

Overfitting the Training Data

Complex model follows training noise

Unstable predictions between and beyond observed cases



Low training error + high validation error = possible overfitting

Diagnosing Overfitting

Observation

Training error is low
Validation error is much higher
The gap remains large

Interpretation

The model fits the training data
The model does not generalize well
Possible overfitting

Use a less flexible model.
Increase regularization.
Collect more representative data.
Remove noisy or irrelevant features.
Check for data leakage.

Underfitting the Training Data

Observation

Training error is high

Validation error is also high

The errors are similar

Interpretation

The model cannot fit the training data well

The model also performs poorly on new data

Possible underfitting

Use a more flexible model.
Create more informative features.
Reduce regularization.
Train the model longer when appropriate.

A Good Training Fit Is Not Enough

We need an evaluation procedure that estimates performance on new data.

How can we evaluate a model without allowing the evaluation data to influence it?



Testing and Validating

The only way to know how well a model will generalize to new cases is to actually try it out on new cases. One way to do that is to put your model in production and monitor how well it performs. This works well, but if your model is horribly bad, your users will complain—not the best idea.

Why Not Test Directly in Production?

Production evaluation provides realistic evidence, but an unreliable model may:

- Give harmful or unfair predictions
- Create a poor user experience
- Cause financial or operational costs

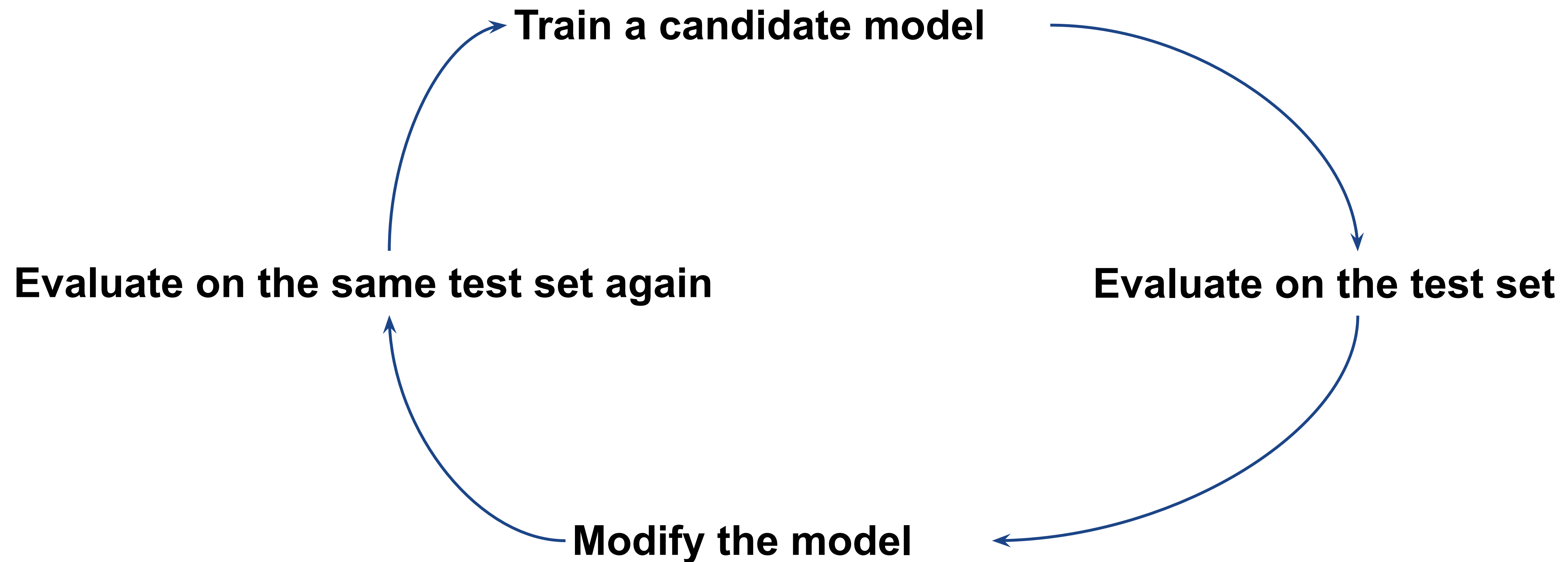
Offline evaluation should identify major problems before deployment.

Training and Test Sets

Training set	Test set
Fit model parameters	Estimate final generalization
Used during development	Kept separate
Can influence the model	Must not influence model choices

A large gap between training and test performance may indicate overfitting.

Repeated Test-Set Use



We never used the test examples to update the model's weights. How could we still overfit the test set?

Training, Validation, and Test Sets

Dataset	Purpose	Can it influence model choices?
Training set	Fit model parameters	Yes
Validation set	Select models and hyperparameters	Yes
Test set	Estimate final generalization	No

Cross-Validation

Round	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
1	Validation	Training	Training	Training	Training
2	Training	Validation	Training	Training	Training
3	Training	Training	Validation	Training	Training
4	Training	Training	Training	Validation	Training
5	Training	Training	Training	Training	Validation

$$\text{CV score} = \frac{1}{K} \sum_{k=1}^K \text{Validation score}_k$$

The test set remains outside cross-validation.

Training and Production Data Mismatch

Training data

Professional web photographs
Clear object and background
Good lighting
Carefully framed

Production data

Mobile phone photographs
Cluttered backgrounds
Variable lighting
Different angles and distances

Diagnosing Data Mismatch with a Train-Dev Set

Training: web images used to fit the model **Train-dev:** web images not used for fitting

Validation: mobile images used for model selection **Test:** mobile images used for final evaluation



Comparison

Training good, train-dev poor

Train-dev good, validation poor

Validation good, test poor

Likely problem

Overfitting

Data mismatch

Validation overfitting or another mismatch

Features Must Exist at Prediction Time.

Feature	Available at prediction time?	Use it?
House size	Yes	Yes
Neighborhood	Yes	Yes
Final sale price	No	No
Future neighborhood price	No	No



Which feature makes the prediction unrealistically easy?

Which Transformations Learn from Data?

Transformation	Learns from data?	Fit using
Convert meters to kilometers	No	Any data
Median imputation	Yes	Training data only
Standardization	Yes	Training data only
One-hot category vocabulary	Yes	Training data only
Logarithmic transformation	Usually no	Can be specified in advance
Feature selection	Yes	Training data only

Data Leakage

- **Target leakage:** A feature directly or indirectly contains the answer.
- **Train-test contamination:** Preprocessing uses the complete dataset.
- **Group leakage:** Records from the same person appear in training and test sets.
- **Temporal leakage:** The model uses future information to predict the past.

Leakage-Safe Pipelines

Split data



Fit preprocessing on training data



Transform validation data



Fit/evaluate model



Test once

Practice: Is this correct?

Impute and standardize the complete dataset



Split the dataset



Train the model



Report test performance

